2D Shapes

Exam code: 0607

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Written by: Jamie Wood
Reviewed by: Dan Finlay
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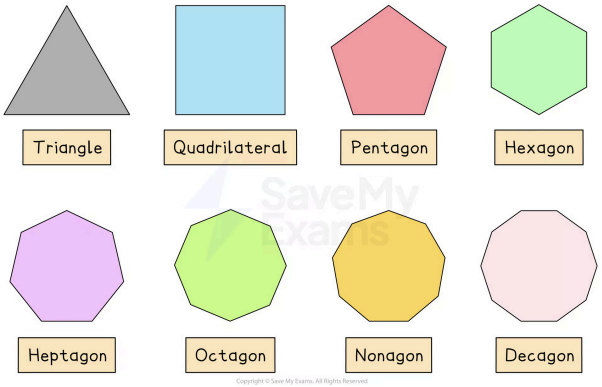
Properties of 2D shapes

What are the names of common 2D shapes?

- You should know the general names of all the 2D **polygons**
  - A **triangle** has 3 sides
  - A **quadrilateral** has 4 sides
  - A **pentagon** has 5 sides
  - A **hexagon** has 6 sides
  - A **heptagon** has 7 sides
  - An **octagon** has 8 sides
  - A **nonagon** has 9 sides
  - A **decagon** has 10 sides
  - A **polygon** is a flat (plane) shape with *n* straight sides
    - A **regular** polygon has all sides the same length and all angles the same size

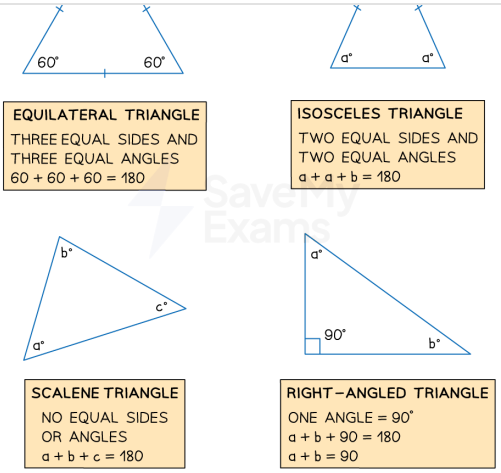


Polygons

What are the names of the different types of triangles?

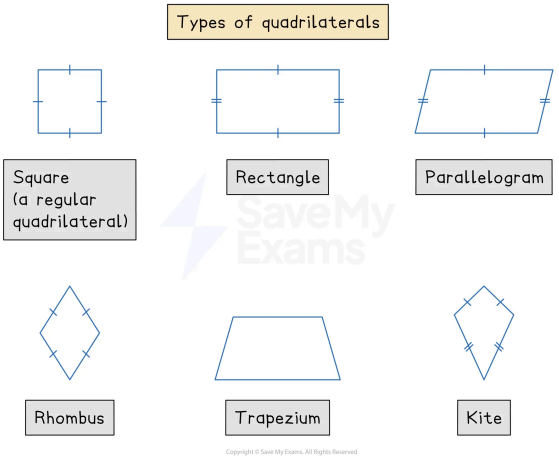
- You should know the names and properties of the different types of **triangles**
  - An **equilateral** triangle has **3 equal sides** and **3 equal angles**
  - An **isosceles** triangle has **2 equal sides** and **2 equal angles**
  - A **right-angled** triangle has **one 90° angle**
  - A **scalene** triangle has **3 sides all of different lengths**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



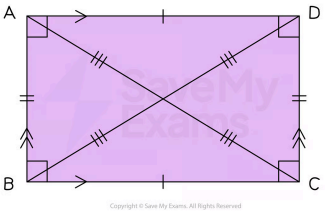
What are the names of the different types of quadrilaterals?

- You should know the names and properties of the different types of **quadrilaterals**
  - These are **squares**, **rectangles**, **parallelograms**, **rhombuses**, **trapeziums** and **kites**



What are the properties of rectangles and squares?

- A **square** is a **rectangle** where all the **sides** have the **same length**
- The table below shows the **properties** of a rectangle and square

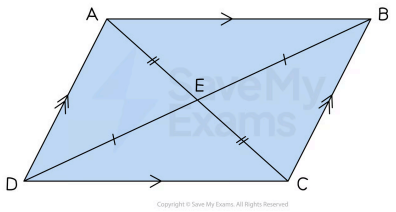


Rectangle

Sides	• Two pairs of equal, opposite sides • Two pairs of parallel sides	• All sides have the same length • Two pairs of parallel sides
Angles	• All angles are 90°	• All angles are 90°
Diagonals	• Both diagonals bisect each other • Diagonals have the same length	• Both diagonals bisect each other • Diagonals have the same length • Diagonals are perpendicular
Symmetries	• Two lines of symmetry • Order of rotational symmetry is two	• Four lines of symmetry • Order of rotational symmetry is four

What are the properties of parallelograms and rhombuses?

- A **rhombus** is a **parallelogram** where all the **sides** have the **same length**
- The table below shows the **properties** of a parallelogram and rhombus



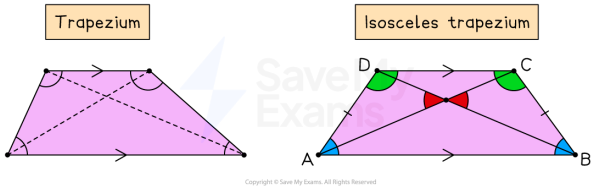
Parallelogram

	Parallelogram	Rhombus
Sides	• Two pairs of equal, opposite sides • Two pairs of parallel sides	• All sides have the same length • Two pairs of parallel sides

	angles • Each pair of adjacent angles add up 180°	adjacent angles add up 180°
Diagonals	• Both diagonals bisect each other	• Both diagonals bisect each other • Diagonals are perpendicular
Symmetries	• No lines of symmetry • Order of rotational symmetry is two	• Two lines of symmetry • Order of rotational symmetry is two

What are the properties of trapeziums?

- The table below shows the **properties** of a trapezium and an isosceles trapezium

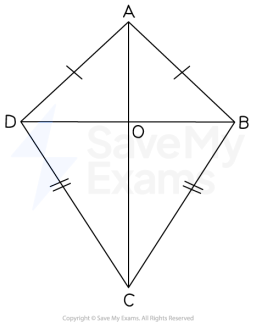


	Trapezium	Isosceles trapezium
Sides	• One pair of parallel sides • The parallel sides have different lengths	• One pair of parallel sides • The parallel sides have different lengths • The non-parallel sides have the same length
Angles	• Two pairs of adjacent angles that add up to 180°	• Two pairs of adjacent angles that add up to 180° • Two pairs of adjacent angles that are equal
Diagonals	• Do not bisect each other	• Do not bisect each other • Diagonals have the same length

	rotational symmetry is one	rotational symmetry is one

What are the properties of kites?

- The table below shows the **properties** of a kite

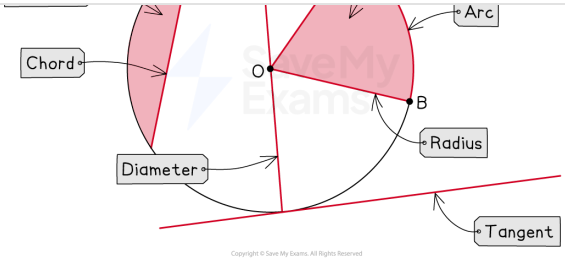


Kite

	Kite
Sides	• Two pairs of equal, adjacent sides • No parallel sides
Angles	• One pair of equal, opposite angles
Diagonals	• One diagonal bisects the other • Diagonals are perpendicular
Symmetries	• One line of symmetry • Order of rotational symmetry is one

What terms related to circles do I need to know?

- **Circles** have several specific terms that you need to be familiar with:
  - A circle's perimeter is called a **circumference**
  - Its line of symmetry is called a **diameter**
  - The line from the centre of the circle to its circumference is called a **radius**
    - The **diameter** is equal to **2 × the radius**
  - A portion of the circumference is called an **arc**
  - A portion of the area, contained between two radii and an arc, is called a **sector**
  - A line between two points on the circumference is called a **chord**
  - The area formed between a chord and an arc is called a **segment**
  - A line which intersects the circumference at one point only, is called a **tangent**



Parts of a circle

- The ratio $\frac{\text{circumference}}{\text{diameter}}$ is equal to π (3.14159...)
- **Circles** have many angle properties and you will need to learn some of them
  - These properties are known as **circle theorems**

Examiner Tips and Tricks

Always double check if a measurement is the **diameter or the radius.** This is a really common error in exams.

Diameter = 2 × Radius.

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